

Deep Learning

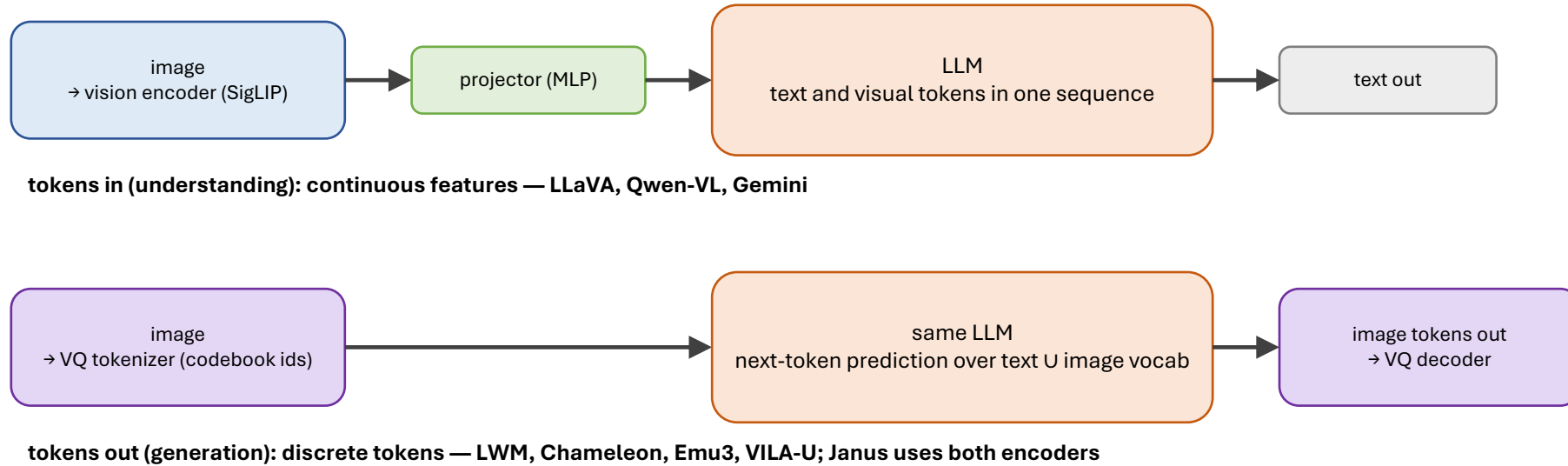
Lecture 9b — Unified multimodal models: tokens in, tokens out
(2026)

Where we are, and today's question

- Lecture 9 covered vision-language transformers that **read** images (VisualBERT, ViLBERT, Flamingo, Mini-GPT-4) and VQ-VAE / DALL-E that **write** them (appendix, slides 96–105)
- Since 2024 the two have merged: one transformer that takes text, images, audio and video **in** and produces text, images and speech **out**
- Two routes: (1) discrete image tokens + next-token prediction; (2) one transformer mixing next-token loss for text with diffusion/flow loss for images

Tokens in: vision-language
models, 2025–26 edition

The recipe has not changed; the details have

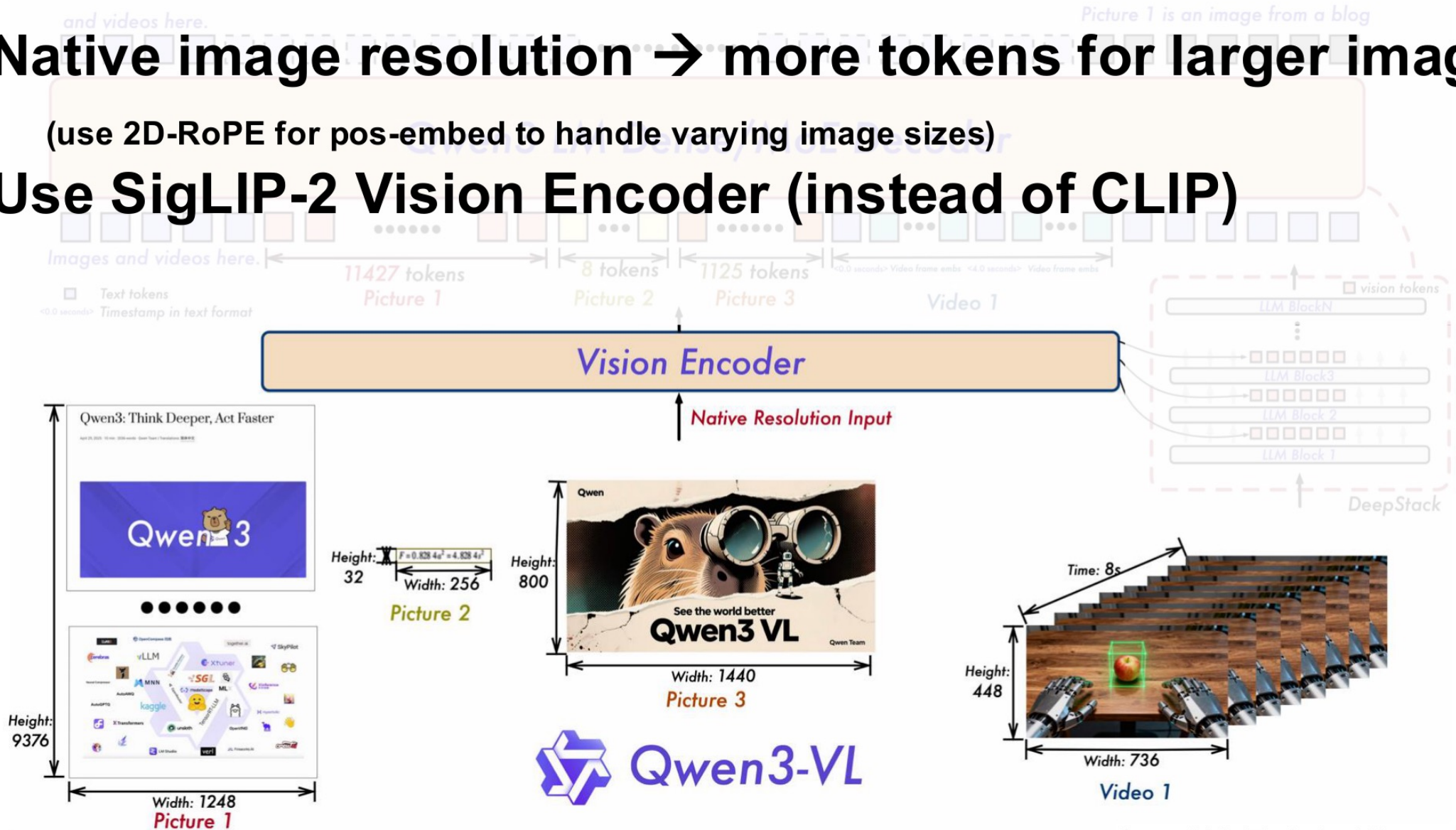


Understanding: vision encoder (CLIP → SigLIP 2) → projector → LLM; visual features enter as continuous tokens (Lecture 9)
2025–26 changes: **native resolution** (variable number of patches, 2D RoPE), video frames with **timestamps** in the text stream, multi-layer visual features, three-stage training (align → multi-task → SFT)
Generation route (bottom): tokenize the image with a VQ codebook and let the *same* LLM emit image ids

1. Native image resolution → more tokens for larger images

(use 2D-RoPE for pos-embed to handle varying image sizes)

1. Use SigLIP-2 Vision Encoder (instead of CLIP)



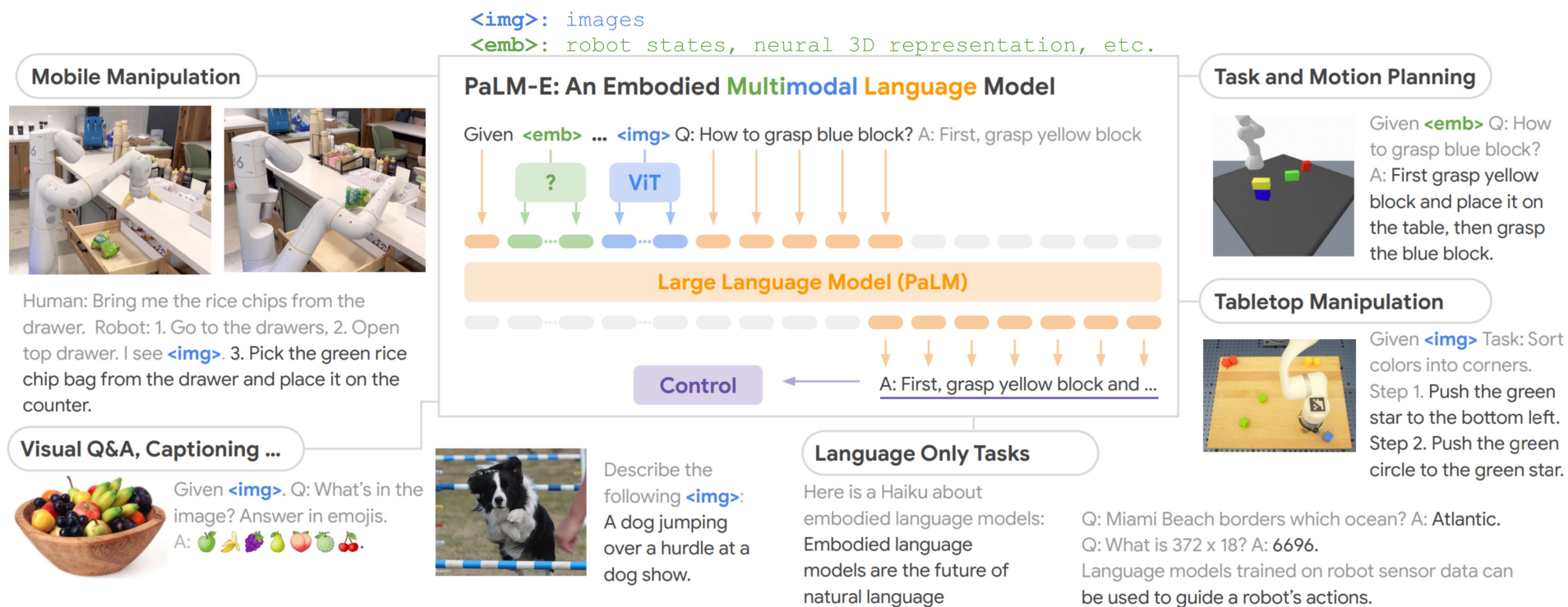
Qwen3-VL Technical Report Dec 2025

What is SOTA in 2026 (so far)?

Gemini widely considered the best proprietary Vision-Language model

- Support flexible modalities (e.g., visual, action)

- Support flexible modalities (e.g., visual, action)



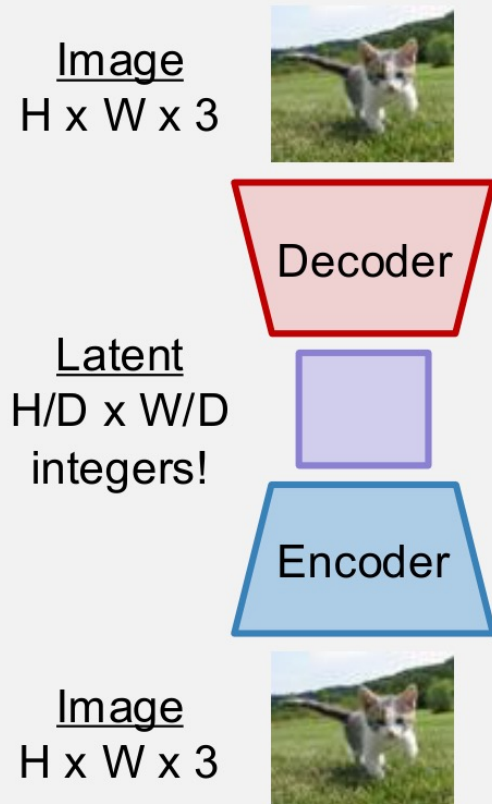
Tokens out: discrete image tokens
and next-token prediction

Why discrete image tokens?

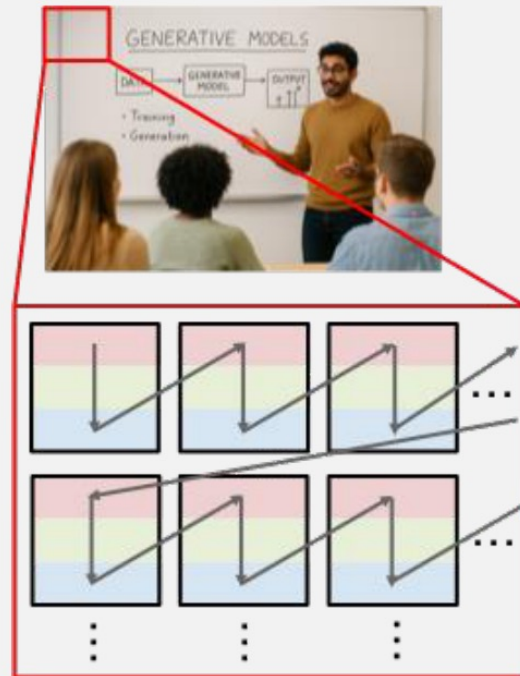
- A VQ-VAE / VQGAN (Lecture 9, slides 96–98) turns an image into a grid of codebook indices — a second vocabulary
- With integer tokens the LLM's machinery is reused unchanged: next-token loss, KV cache, sampling, the same weights for text and images
- Trade-off: CLIP-style features are better for understanding (semantics), VQ tokens are needed for generation (pixels) — the tension Janus resolves
- Large World Model (Liu et al., 2024): VQGAN tokens for images and video + text in one transformer with a 1M-token context

Autoregressive Models Strike Back

Train **encoder** + **decoder**
to convert images to
discrete latents



Train **autoregressive model**
to model sequences of
discrete latents



Sample discrete latents from
the **autoregressive model**,
pass to **decoder** to get an
image

van den Oord et al, "Neural Discrete Representation Learning", NeurIPS 2017
Razavi et al, "Generating Diverse High-Fidelity Images with VQ-VAE-2", NeurIPS 2019
Esser et al, "Taming Transformers for High-Resolution Image Synthesis", CVPR 2021
Yu et al, "Scaling Autoregressive Models for Content-Rich Text-to-Image Generation", arXiv 2022

Next-token image generation grew up: Chameleon, Emu3, VAR

- **Chameleon** (Meta, May 2024): early fusion — text and VQ image tokens in one vocabulary, one transformer trained from scratch on both
- **Emu3** (BAAI, Sep 2024): next-token prediction only, for images, video and text — competitive with SDXL and LLaVA-1.6 in the authors' evaluation
- **VAR** (Tian et al., NeurIPS 2024 best paper): predict the image coarse-to-fine, **one scale at a time** instead of raster order; GPT-style AR beat diffusion transformers on ImageNet FID
- Lesson: the tokenizer and the prediction order matter as much as the transformer

VILA-U: Unified Visual Language Model

VILA-U unifies Video, Image and Language understanding and generation within one single autoregressive model.

- Natively support visual generation in the LLM inference pipeline
- Achieve near SOTA performance with discrete visual features in visual understanding tasks Success due to two main factors:
- A Unified Vision Tower that aligns discrete visual tokens with textual inputs
- A Unified Token-in and Token-out training and inference pipeline



Prompt: Image 1: <img1>\n Image 2: <img2>\n

What is the common thing and the difference between image 1 and image 2?

VILA-U(Ours): The common thing in both images is the presence of a panda bear. However, the difference between image 1 and image 2 is that the panda bear in image 1 is lying down on a rock, while the panda bear in image 2 is lying down on a log.



Happy dreamy owl monster sitting on a tree branch, colorful glittering particles, forest background, detailed feathers



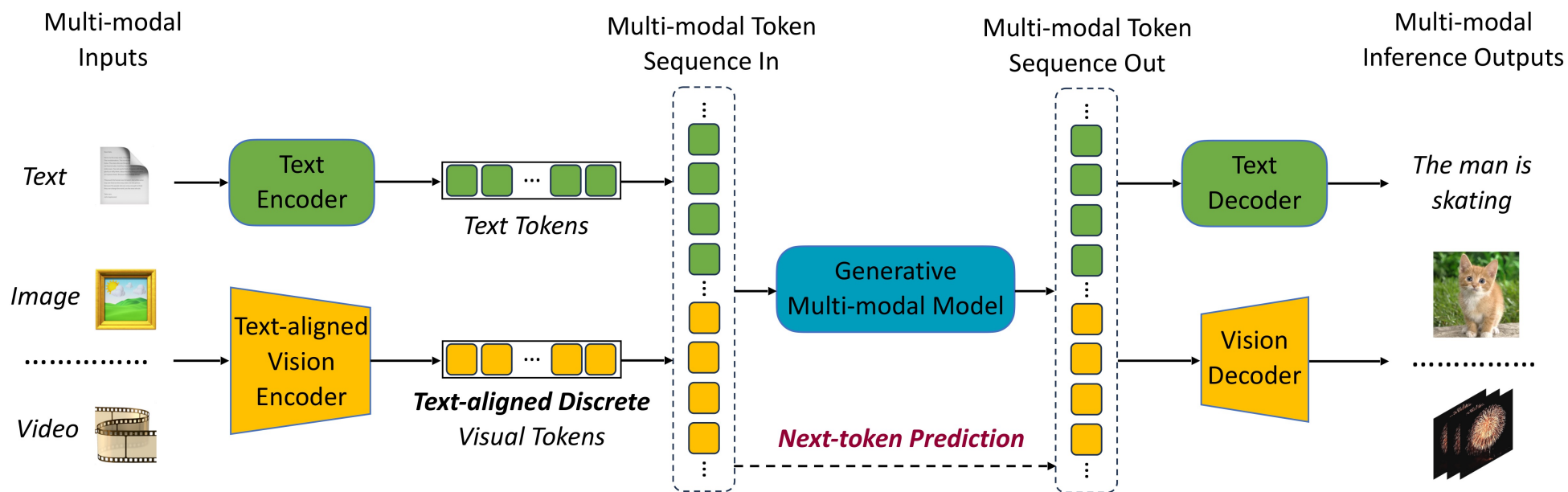
A snowy mountain



An extreme close-up of an gray-haired man with a beard in his 60s

VILA-U: Unified Multi-modal Pre-training

Unified Multi-modal Generative Pre-Training: Token in and Token out



- Training: Multi-modal inputs -- encoder --> discrete tokens, then LM loss can be applied on any tokens!
- Inference: Multi-modal tokens --- decoder ---> text / image / video

VILA-U: Understanding Results

Performance on Image Understanding Benchmarks

Method	LLM	Visual Token	Res.	VQAv2	GQA	TextVQA	POPE	MME	SEED	MM-Vet
LLaVA-1.5	Vicuna-1.5-7B	Continuous	336	78.5*	62.0*	58.2	85.9	1510.7	58.6	30.5
VILA	LLaMA-2-7B	Continuous	336	79.9*	62.3*	64.4	85.5	1533.0	61.1	34.9
Unified-IO 2	6.8B from scratch	Continuous	384	79.4*	—	—	87.7	—	61.8	—
InstructBLIP	Vicuna-7B	Continuous	224	—	49.2	50.1	—	—	53.4	26.2
IDEFICS-9B	LLaMA-7B	Continuous	224	50.9	38.4	25.9	—	—	—	—
Emu	LLaMA-13B	Continuous	224	52.0	—	—	—	—	—	—
LaVIT	LLaMA-7B	Continuous	224	66.0	46.8	—	—	—	—	—
DreamLLM	Vicuna-7B	Continuous	224	72.9*	—	41.8	—	—	—	36.6
Video-LLaVIT	LLaMA-2-7B	Continuous	224	80.2*	63.6*	—	—	1581.5	64.4	35.0
CM3Leon-7B	7B from scratch	Discrete	256	47.6	—	—	—	—	—	—
LWM	LLaMA-2-7B	Discrete	256	55.8	44.8	18.8	75.2	—	—	9.6
Show-o	Phi-1.5-1.3B	Discrete	256	59.3*	48.7*	—	73.8	948.4	—	—
Ours	LLaMA-2-7B	Discrete	256	75.3*	58.3*	48.3	83.9	1336.2	56.3	27.7
Ours	LLaMA-2-7B	Discrete	384	79.4*	60.8*	60.8	85.8	1401.8	59.0	33.5

- Our method achieves far better performance than other autoregressive methods utilizing discrete visual tokens
- For the first time, discrete visual token VLM can match continuous token!

VILA-U: Generation results



A black dog



An oil painting of a garden where every flower is in full bloom, showcasing a rainbow of colors



An elephant walking under A cube made of denim
The sea



A realistic landscape shot of the Northern Lights dancing over a snowy mountain range in Iceland



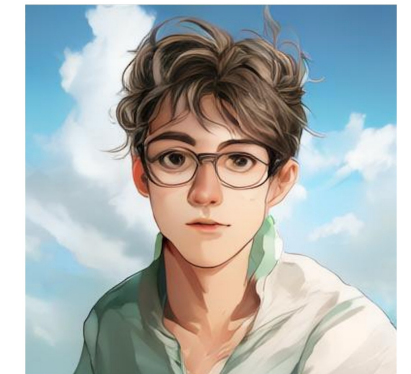
An extreme close-up of an gray-haired man with a beard in his 60s



Knolling of a drawing tools for A man looks up at the starry sky, lonely and ethereal



Drone view of waves crashing middle with sky color background painter against the rugged cliffs in Big Sur wearing eye glasses, it's super



a handsome 24 years old boy in the background painter

Janus: decouple the two visual encoders, keep one transformer



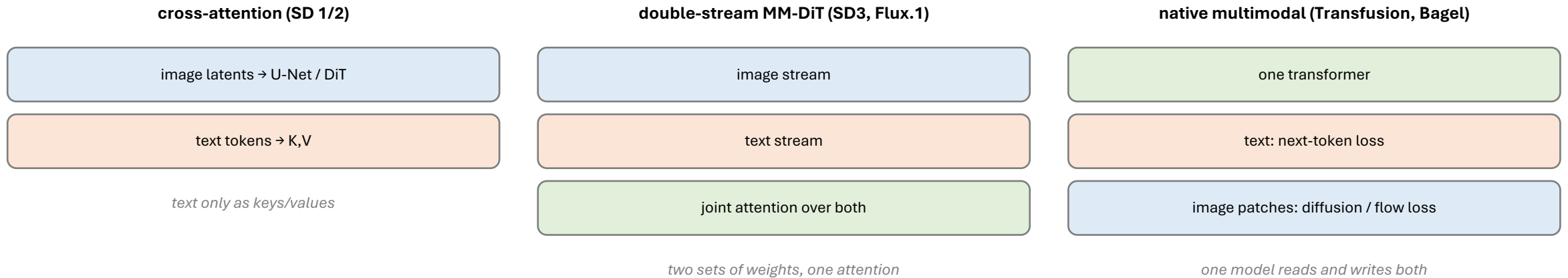
decoupling removes the conflict between semantic features (understanding) and pixel detail (generation)

Understanding and generation need different information (semantics vs pixel detail); one encoder compromises both
Janus (DeepSeek, Oct 2024): a SigLIP-style encoder for understanding, a VQ tokenizer for generation, one autoregressive transformer that also emits image tokens; the authors report it surpasses previous unified models and matches task-specific ones

Janus-Pro (Jan 2025): scaled data and training, 7B; **JanusFlow** (Nov 2024): the generation branch becomes rectified flow (Lecture 10b)

One model, two objectives: next-token for text, diffusion/flow for images

Mixing objectives in one transformer



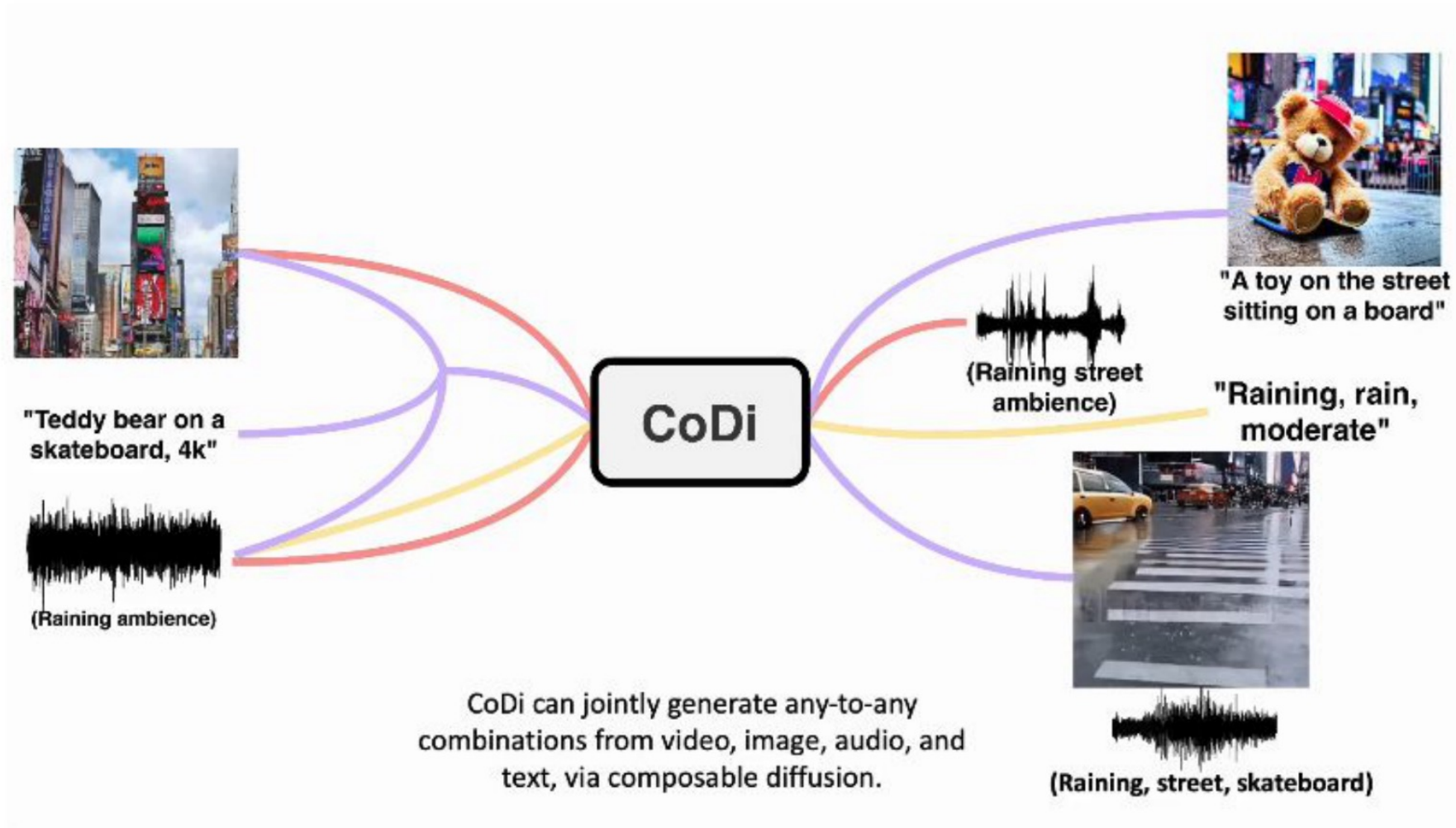
Transfusion (Meta, Aug 2024): one transformer; next-token loss on text tokens, diffusion loss on continuous image patches; markers delimit image spans

Show-o (Aug 2024): AR text + *discrete* diffusion over image tokens; **Bagel** (ByteDance, May 2025): mixture-of-transformer-experts, 7B active, trained on interleaved video/web data — understanding, generation and editing

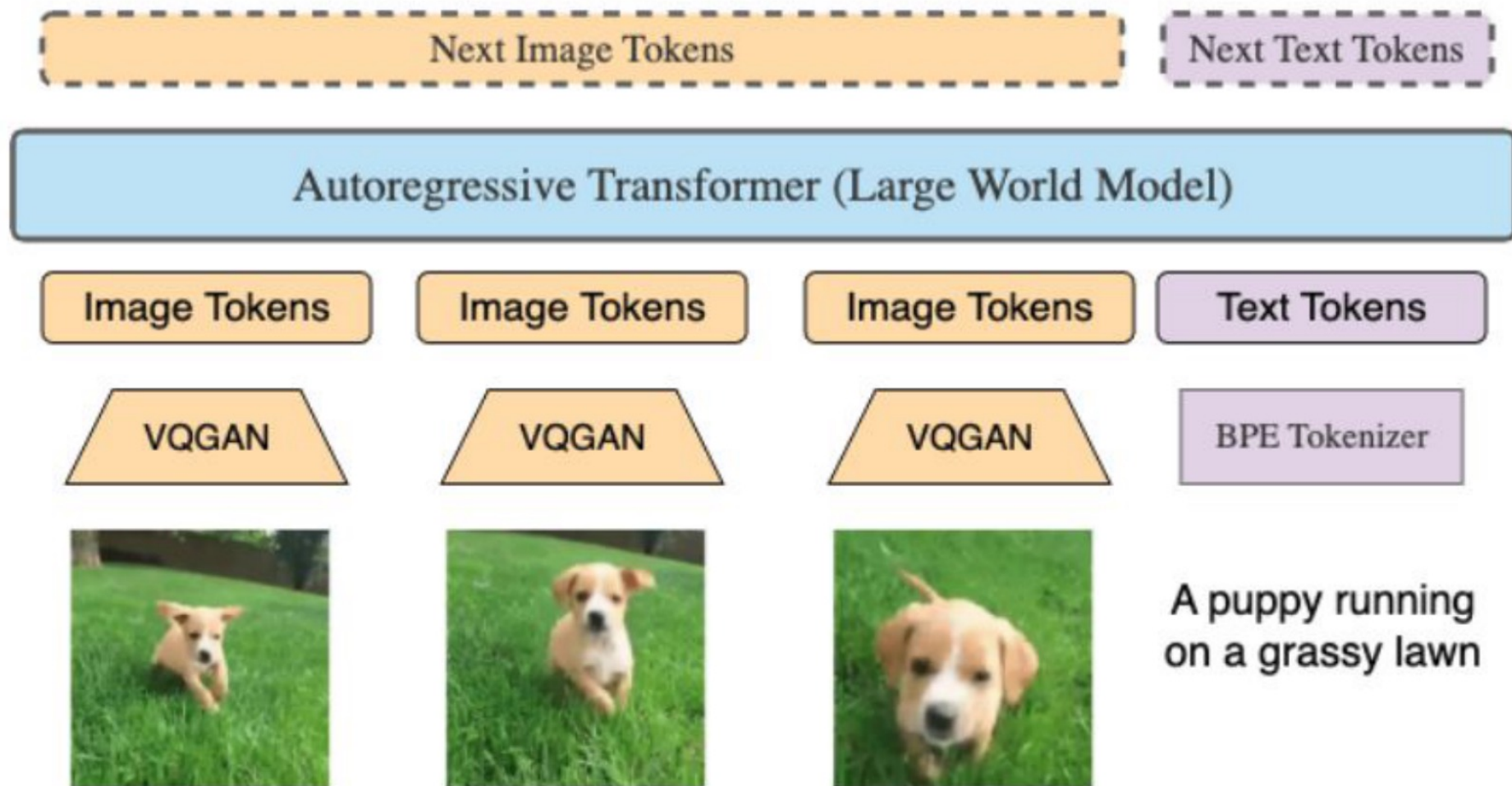
Products: **GPT-4o native image generation** (Mar 2025), Gemini's **Nano Banana** (Aug 2025) and Nano Banana Pro (Nov 2025) — editing with the chat model's world knowledge and multi-turn context; internals not public

Omni models: any modality in,
text and speech out

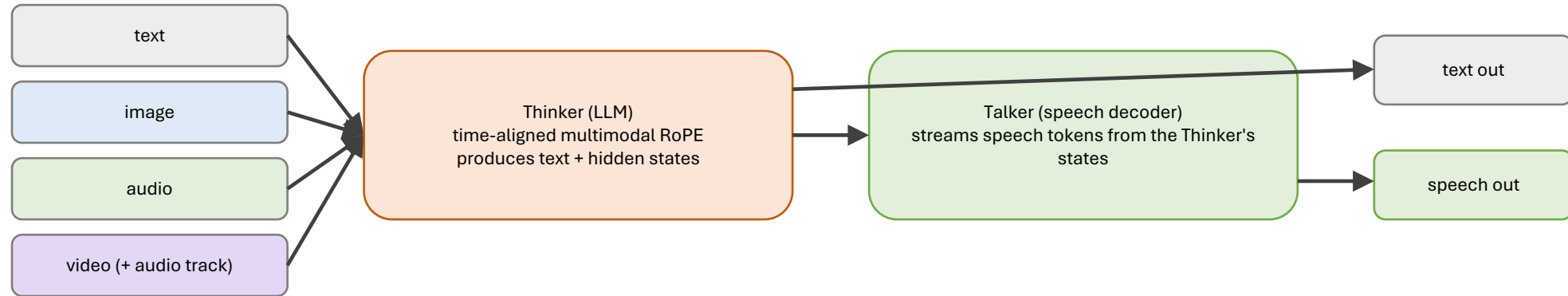
CoDi: Any-to-Any Generation



LWM



Any-to-any in 2025: Thinker–Talker



Qwen2.5-Omni (Mar 2025) and Qwen3-Omni (Sep 2025): any-to-any with a two-part design; GPT-4o (2024) and Gemini Live use single end-to-end models (details not public)

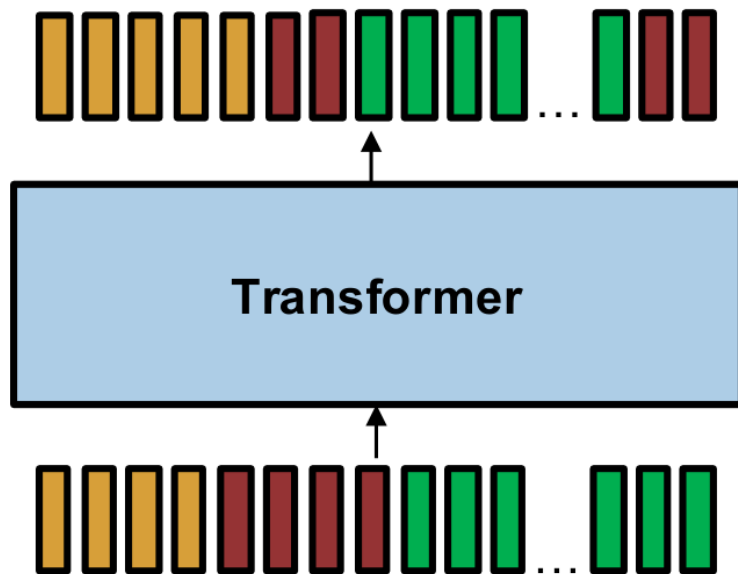
Qwen2.5-Omni (Mar 2025): a Thinker LLM consumes text, image, audio and video with time-aligned multimodal RoPE; a Talker streams speech from the Thinker's hidden states

Qwen3-Omni (Sep 2025): 119 languages, long audio input, real-time speech output, a thinking mode

CS231n's closing note (May 2026): omni models started with GPT-4o (2024); new omni models from Thinking Machines and Google in May 2026

Final Note: Omni Models (Beyond Vision + Language)

Started with GPT-4o in 2024 Recently, Thinking Machines + Gemini have introduced their own models as well



Train a model to unput and output in text, audio, and video.

[Thinking Machines post from 2 weeks ago](#)
[Google blog post from 3 hours ago](#)

Summary and where this goes in the course

- Understanding = images as continuous tokens in (CLIP/SigLIP + projector); generation = discrete tokens out (VQ) or a diffusion/flow head inside the transformer
- Unified models resolve the encoder tension by decoupling (Janus) or by mixing objectives (Transfusion, Bagel); products (GPT-4o images, Nano Banana) show the payoff: editing with world knowledge and context
- Omni models add audio and speech with the same recipe; robotics (RT-2, $\pi 0$) treats actions as one more output modality (Lecture 11a)
- Suggested placement: after Lecture 9's Mini-GPT-4 slides; move the VQ-VAE/dVAE appendix into the main flow before this section